

Seller and Inventory Management System

1. Create Database

```
CREATE DATABASE SellerInventoryDB;  
USE SellerInventoryDB;
```

2. Create Seller Table

```
CREATE TABLE Seller (  
    seller_id INT PRIMARY KEY,  
    seller_name VARCHAR(100),  
    email VARCHAR(100),  
    phone VARCHAR(15)  
);
```

3. Create Product Table

```
CREATE TABLE Product (  
    product_id INT PRIMARY KEY,  
    product_name VARCHAR(100),  
    price DECIMAL(10,2),  
    seller_id INT,  
    FOREIGN KEY (seller_id) REFERENCES Seller(seller_id)  
);
```

4. Create Inventory Table

```
CREATE TABLE Inventory (  
    inventory_id INT PRIMARY KEY,  
    product_id INT,  
    stock_quantity INT,  
    status VARCHAR(20),  
    FOREIGN KEY (product_id) REFERENCES Product(product_id)  
);
```

5. Insert Seller Details

```
INSERT INTO Seller VALUES  
(1, 'Ravi', 'ravi@gmail.com', '9876543210'),  
(2, 'Priya', 'priya@gmail.com', '9876543211'),  
(3, 'Arun', 'arun@gmail.com', '9876543212');
```

6. Insert Product Details

```
INSERT INTO Product VALUES  
(101, 'Laptop', 55000.00, 1),  
(102, 'Mouse', 800.00, 1),  
(103, 'Keyboard', 1500.00, 2),  
(104, 'Headphones', 2500.00, 3);
```

7. Insert Inventory Details

```
INSERT INTO Inventory VALUES  
(1, 101, 10, 'Available'),  
(2, 102, 0, 'Unavailable'),  
(3, 103, 25, 'Available'),  
(4, 104, 0, 'Unavailable');
```

8. Display All Sellers

```
SELECT * FROM Seller;
```

9. Display All Products

```
SELECT * FROM Product;
```

10. Display Inventory Status

```
SELECT * FROM Inventory;
```

11. Find Available Products

```
SELECT  
    Product.product_name,  
    Inventory.stock_quantity,  
    Inventory.status  
FROM Product  
JOIN Inventory  
ON Product.product_id = Inventory.product_id  
WHERE Inventory.status = 'Available';
```

12. Find Unavailable Products

```
SELECT  
    Product.product_name,  
    Inventory.stock_quantity,  
    Inventory.status  
FROM Product  
JOIN Inventory  
ON Product.product_id = Inventory.product_id  
WHERE Inventory.status = 'Unavailable';
```

13. Seller + Product Information

```
SELECT  
    Seller.seller_name,  
    Product.product_name,  
    Product.price  
FROM Seller  
JOIN Product  
ON Seller.seller_id = Product.seller_id;
```

14. Inventory Status Report

```
SELECT
    Seller.seller_name,
    Product.product_name,
    Inventory.stock_quantity,
    Inventory.status
FROM Seller
JOIN Product
ON Seller.seller_id = Product.seller_id
JOIN Inventory
ON Product.product_id = Inventory.product_id;
```

What this project covers

Requirement	SQL implementation
Create Seller table	<code>Seller</code>
Create Inventory table	<code>Inventory</code>
Create Product table	<code>Product</code>
Establish relationships	Foreign Keys
Maintain seller products	<code>Seller → Product</code>
Track available products	<code>status = Available</code>
Track unavailable products	<code>status = Unavailable</code>
Generate inventory report	<code>JOIN</code> query

Relationship:

`Seller (1) → Product (Many) → Inventory (1)`